

- 3 (a) With reference to molecular kinetic energy and molecular potential energy, explain what is meant by the internal energy of an ideal gas.

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 [2]

- (b) A sample of an ideal gas is initially in state A, at a pressure of $2.0 \times 10^5 \text{ Pa}$ and with a volume of 0.016 m^3 , as shown in Fig. 3.1.

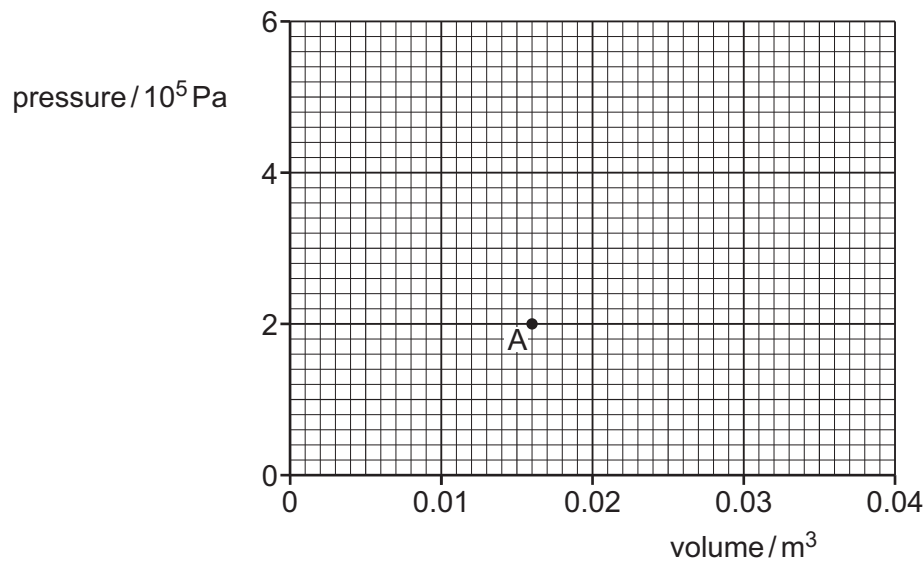


Fig. 3.1

In state A, the temperature of the gas is 400 K.

The gas undergoes two successive changes X and Y.

In change X, it is heated at constant volume to a pressure of $4.0 \times 10^5 \text{ Pa}$. At the end of change X, the gas is in state B.

In change Y, it is then allowed to expand at constant temperature back to its original pressure. At the end of change Y, the gas is in state C.

